

Frontend Developer Guide (React + MUI + React Query)

The entire frontend framework and UX flow.

We implemented the following major components:

- React with Vite for optimized SPA
- Material UI (MUI) as primary design system
- React Router v6 for route management
- React Query for data fetching and caching
- Axios (apiClient.js) with JWT interceptor and global 401/403 handler
- Role-based routing (ProtectedUserRoute, ProtectedManagerRoute, ProtectedAdminRoute)
- Local Storage for token and role persistence
- Sidebar navigation across dashboards (User, Manager, Admin)
- Global token expiry auto-redirect to login

Frontend Directory Structure

- client/src/ pages/ #
User views components/ # Shared components bookings, hotels, etc.)
- manager/pages/ #
Manager views utils/ # Token and date helpers wrappers) App.jsx # Route structure Core Frontend Flows
- admin/pages/ #
Admin views layouts/ # Sidebar layouts services/ # Axios services (auth, hooks

Core Frontend Flows

Root entry User Portal:

- Explore hotels (no login required) • Book rooms → prompts login → creates booking with status DRAFT • View/Edit bookings in dashboard • Proceed to pay → Pending → Paid • View status updates and cancel bookings

Manager Portal:

- Register hotel and rooms
- Manage room availability and pricing
- View all customer bookings
- Update booking statuses (Checked-in / Completed)
- Prevent editing hotels with active bookings

Admin Portal:

- Login-only access
- Approve or deactivate hotels
- View all users, managers, bookings, and payments
- Monitor commissions, revenue, and analytics (future phase)

Global UI Rules

- All dashboards use MUI Drawer (Sidebar-left layout)
- Common color theme for consistency
- Snack bar + Alert for feedback
- MUI Skeletons for loading states
- Auto token expiry → redirect to correct login portal
- 3 separate logins: /login (User), /manager/login, /admin/login